

AWS Cloud Computing Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Course Overview

This syllabus lists every module of the course in teaching order, with the topics covered in each. It is generated from the course page, so the two cannot disagree.

Module 1: Cloud Foundations & AWS Account Setup

Schedule: Week 1

Cloud computing concepts (IaaS / PaaS / SaaS, on-prem vs cloud trade-offs, shared responsibility model), the AWS global infrastructure (Regions, AZs, Edge locations, Wavelength zones), account creation, billing alarms, the AWS Free Tier, and IAM basics. By the end of week 1 every student has a personal AWS sandbox account, MFA enabled, a budget alarm at ₹1,000, and an IAM user with programmatic access. We deliberately spend a full session on cost — every horror story you have heard about a runaway AWS bill starts with what we cover this week.

Topics

- Cloud computing models — IaaS / PaaS / SaaS
- AWS global infrastructure — Regions, AZs, Edge locations
- Shared responsibility model
- Account setup, MFA, root vs IAM users
- Billing alarms, AWS Budgets, Cost Explorer
- AWS Free Tier — what's free and for how long
- AWS CLI v2 setup and named profiles

Module 2: IAM, Identity Federation & Security Baseline

Schedule: Week 2

Identity is the foundation of cloud security. Learn IAM users, groups, roles, and policies (managed, customer-managed, inline) — and the design principles that separate hobbyist accounts from production-grade ones. Cover IAM Identity Center (the successor to AWS SSO) for workforce identity, OIDC federation for CI/CD pipelines, and AWS Organizations + SCPs for multi-account governance. We finish the module by hardening the sandbox account using the AWS Security Hub baseline.

Topics

- IAM users, groups, roles, policies
- Policy evaluation logic — explicit deny, boundaries, conditions
- IAM roles for EC2, Lambda, and cross-account access
- IAM Identity Center for workforce SSO
- OIDC federation for GitHub Actions / Jenkins
- AWS Organizations and Service Control Policies
- Security Hub, GuardDuty, AWS Config baseline

Module 3: Networking — VPC, Subnets, Routing & Isolation

Schedule: Week 3

Cloud networking is where most certification candidates lose marks and most engineers lose production weekends. We build a VPC from scratch — public and private subnets across two AZs, a NAT Gateway, an Internet Gateway, route tables, security groups, and Network ACLs — and prove the design with a working three-tier application (ALB → EC2 → RDS).

CIDR planning gets its own session, because a VPC range chosen badly in week one is the thing you cannot change later without rebuilding. We size subnets against realistic growth, reserve space for future AZs, and check the plan against the on-prem ranges an enterprise would need to peer with.

Topics

- VPC, subnets, CIDR planning and address exhaustion
- Internet Gateway, NAT Gateway, NAT Instance
- Route tables and the implicit local route
- Security groups vs Network ACLs — stateful vs stateless
- Public / private / isolated subnet tiers
- VPC Flow Logs for traffic troubleshooting
- Building a two-AZ three-tier VPC end to end

Module 4: Hybrid Connectivity, Endpoints & Route 53

Schedule: Week 4

The second half of AWS networking — connecting a VPC to other VPCs, to on-premises, and to AWS services without traversing the public internet. VPC Peering and its non-transitive limitation, Transit Gateway as the hub-and-spoke answer, Gateway and Interface endpoints, PrivateLink for exposing your own services, and Direct Connect versus Site-to-Site VPN.

Route 53 closes the module: hosted zones, record types, and the routing policies (weighted, latency, failover, geolocation) that appear in both the SAA-C03 exam and every real disaster-recovery design.

Topics

- VPC Peering and why it is not transitive
- Transit Gateway — hub-and-spoke at scale
- VPC endpoints — Gateway (S3, DynamoDB) and Interface
- AWS PrivateLink for private service exposure
- Site-to-Site VPN and Direct Connect trade-offs
- Route 53 hosted zones and record types
- Routing policies — weighted, latency, failover, geolocation
- Health checks and DNS-level failover

Module 5: Compute — EC2, AMIs & Auto Scaling

Schedule: Week 5

The compute service everything else is measured against. EC2 instance families and what each one is actually for, the four pricing models (on-demand, spot, savings plans, reserved) and how to combine them, storage-optimised versus compute-optimised selection, AMI lifecycle and EC2 Image Builder, and user data for bootstrapping.

Then resilience: Auto Scaling groups with launch templates, target tracking and step scaling policies, health checks, and an Application Load Balancer in front. The lab ends with an application that survives an instance being terminated by hand.

Topics

- EC2 instance families and sizing
- Pricing models — on-demand, spot, savings plans, reserved
- Spot interruption handling and mixed-instance policies
- AMIs, snapshots, EC2 Image Builder
- User data, instance metadata and IMDSv2
- Launch templates and Auto Scaling groups
- Target tracking, step and scheduled scaling
- Application Load Balancer, target groups, health checks
- Elastic Beanstalk overview

Module 6: Serverless — Lambda, API Gateway & Event-Driven Design

Schedule: Week 6

Serverless as an architecture, not a trick. Lambda in depth — runtimes (Java 21, Python 3.13, Node 22), layers, memory-to-CPU coupling, timeouts, concurrency limits, and the cold-start behaviour that decides whether serverless suits a workload at all. Then the services Lambda is glued to: API Gateway REST and HTTP APIs, Step Functions for orchestration, EventBridge for routing, and SQS / SNS for decoupling.

The lab builds a small event-driven service end to end, so the trade-off against the EC2 design from the previous module is something you have measured rather than read about.

Topics

- Lambda runtimes, memory / CPU coupling, timeouts
- Layers, environment variables, versions and aliases
- Cold starts, provisioned concurrency, SnapStart
- API Gateway — REST vs HTTP APIs, authorisers, throttling
- Step Functions — standard vs express workflows
- EventBridge rules, buses and schemas
- SQS and SNS for decoupling and fan-out
- Building an event-driven service end to end

Module 7: Containers on AWS — ECR, ECS, Fargate & EKS

Schedule: Week 7

Where most Pune cloud roles actually spend their time. Push images to Amazon ECR with scanning enabled, then run them two ways: Amazon ECS with Fargate (task definitions, services, service discovery, autoscaling) and Amazon EKS (managed node groups, Fargate profiles, the AWS Load Balancer Controller, IRSA for pod-level IAM).

We are explicit about when each is the right answer — ECS for a team that wants AWS to hold the abstraction, EKS for a team that needs the Kubernetes ecosystem or portability. You build a Fargate-only EKS cluster you can run for under ₹100 a day.

Topics

- Amazon ECR — repositories, lifecycle policies, image scanning
- ECS task definitions, services and clusters
- Fargate versus EC2 launch types
- ECS service autoscaling and service discovery
- EKS cluster provisioning with eksctl
- Managed node groups versus Fargate profiles
- IAM Roles for Service Accounts (IRSA)
- AWS Load Balancer Controller and ingress
- Choosing between ECS and EKS

Module 8: Storage — S3, EBS, EFS & Data Lifecycle

Schedule: Week 8

S3 in the depth the exam and the job both require: storage classes and the retrieval cost that makes Glacier cheap only if you never read it, lifecycle rules, versioning and MFA delete, replication (same-region and cross-region), encryption modes, signed URLs, static website hosting, and S3 Object Lambda.

Block and file storage follow — EBS volume types and the gp3 baseline-versus-provisioned distinction, snapshots and fast snapshot restore, EFS for shared POSIX filesystems, and an overview of FSx. The module closes on choosing correctly between the three under a stated cost and latency budget.

Topics

- S3 storage classes and retrieval cost trade-offs
- Lifecycle policies, Intelligent-Tiering, expiry
- Versioning, MFA delete and Object Lock
- Same-region and cross-region replication
- S3 encryption — SSE-S3, SSE-KMS, SSE-C
- Signed URLs, bucket policies and Block Public Access
- S3 static hosting and CloudFront origin patterns
- EBS volume types — gp3, io2, st1, sc1
- Snapshots, fast snapshot restore, multi-attach
- EFS and FSx for Windows / Lustre overview

Module 9: Databases — RDS, Aurora, DynamoDB & Caching

Schedule: Week 9

Managed data services and the design decisions that outlast them. RDS engines, Multi-AZ for availability versus read replicas for scale (a distinction interviewers reliably probe), parameter groups, backups and point-in-time recovery. Aurora goes deeper — the shared storage layer, Serverless v2 scaling, and Aurora Global for cross-region.

DynamoDB is taught as a modelling discipline rather than a key-value store: partition key design, hot partitions, single-table design, GSIs and LSIs, on-demand versus provisioned capacity, and Streams. ElastiCache and OpenSearch close the module.

Topics

- RDS engines, subnet groups, parameter groups
- Multi-AZ for availability vs read replicas for scale
- Automated backups and point-in-time recovery
- Aurora architecture and the shared storage layer
- Aurora Serverless v2 and Aurora Global
- DynamoDB partition key design and hot partitions
- Single-table design, GSIs and LSIs
- On-demand vs provisioned capacity and autoscaling
- DynamoDB Streams and TTL
- ElastiCache (Redis) and OpenSearch basics

Module 10: Infrastructure as Code — Terraform & AWS CDK

Schedule: Week 10

Modern cloud is code, not clicks. Terraform 1.7+ first — providers, resources, variables and outputs, remote state in S3 with DynamoDB locking, modules, workspaces, plan as a review artefact, and the import workflow for adopting resources someone created by hand. Then AWS CDK in TypeScript — constructs, stacks, and how synthesis produces CloudFormation underneath.

We teach both because Pune teams use both and you will be asked to read each. CloudFormation itself is covered at the level needed to debug what CDK emits: stacks, change sets, and drift detection.

Topics

- Terraform providers, resources, variables, outputs
- Remote state in S3 with DynamoDB locking
- Modules, workspaces and environment separation
- terraform plan as a code-review artefact
- Importing hand-created resources
- AWS CDK in TypeScript — constructs and stacks
- CDK synthesis and the CloudFormation output
- CloudFormation stacks, change sets, drift detection

- Choosing Terraform vs CDK on a real team

Module 11: CI/CD Pipelines & Deployment Strategies

Schedule: Week 10

Wiring the code you write to the infrastructure you defined. GitHub Actions deploying via OIDC federation — no static access keys anywhere, which is now the expected answer in interviews — plus AWS-native CodePipeline, CodeBuild and CodeDeploy for shops standardised on AWS tooling.

Deployment strategy is the second half: rolling, blue/green with ALB target-group swaps, and canary releases with CodeDeploy traffic shifting. Every pattern is paired with its rollback path, because a deployment strategy without a rollback plan is just optimism.

Topics

- GitHub Actions with OIDC — no long-lived keys
- Build, test and artefact stages
- CodePipeline, CodeBuild, CodeDeploy
- Rolling, blue/green and canary deployments
- ALB target-group swaps for blue/green
- CodeDeploy traffic shifting and automatic rollback
- Secrets in pipelines — OIDC, Secrets Manager, masked outputs
- Environment promotion — dev, staging, production

Module 12: Observability — CloudWatch, X-Ray & OpenTelemetry

Schedule: Week 11

Observability the way SREs actually configure it, not the way marketing pages describe it. CloudWatch metrics and custom metrics, structured JSON logging queried with Logs Insights, alarms with sensible thresholds feeding SNS, composite alarms to suppress noise, and dashboards that a person on call can read at 3am.

Distributed tracing follows with X-Ray and the AWS Distro for OpenTelemetry, plus a small Prometheus and Grafana stack on EKS for the cloud-native pattern you will meet in container-first teams.

Topics

- CloudWatch metrics, custom metrics and dimensions
- Structured JSON logging and Logs Insights queries
- Alarms, composite alarms and alarm fatigue
- Dashboards built for on-call readability
- AWS X-Ray — segments, subsegments, service maps
- AWS Distro for OpenTelemetry
- Prometheus and Grafana on EKS
- SNS, SQS and EventBridge for alert routing

Module 13: Security, Encryption & Well-Architected Review

Schedule: Week 12

The SAA-C03 exam weights security heavily, and so do real Pune hiring panels. Cover KMS in depth (CMKs, grants, envelope encryption, key rotation), AWS Secrets Manager and Parameter Store, ACM for TLS certificates, AWS WAF and AWS Shield (Standard and Advanced), Macie for data classification, and the Well-Architected Tool's six pillars.

We finish by reviewing each student's capstone architecture against the Well-Architected lens — a written review, in the format AWS partners actually produce.

Topics

- KMS — CMKs, grants, envelope encryption, key rotation

- Secrets Manager vs SSM Parameter Store
- ACM and TLS termination at ALB / CloudFront
- WAF rules, Shield Standard vs Advanced
- GuardDuty, Macie, Inspector overview
- Well-Architected Framework — six pillars
- Compliance — PCI DSS, HIPAA, SOC 2 baseline mapping
- Writing a Well-Architected review document

Module 14: Cost Optimisation & FinOps Practice

Schedule: Week 12

The skill that gets a cloud engineer promoted and the one most syllabi omit. Reading a bill properly with Cost Explorer and the Cost and Usage Report, allocating spend with tagging strategies and cost categories, and setting Budgets with actions that actually stop runaway resources rather than merely emailing about them.

Then the levers: right-sizing from CloudWatch data, Savings Plans versus Reserved Instances, spot for interruptible work, S3 lifecycle and storage-class analysis, and the standard waste list — idle load balancers, unattached EBS volumes, orphaned snapshots, over-provisioned RDS.

Topics

- Cost Explorer and the Cost and Usage Report
- Tagging strategy and cost allocation tags
- AWS Budgets with budget actions
- Right-sizing from CloudWatch utilisation data
- Savings Plans vs Reserved Instances vs spot
- Storage-class analysis and S3 lifecycle savings
- Finding waste — idle ALBs, unattached EBS, orphaned snapshots
- Building a monthly FinOps review habit

Module 15: Generative AI on AWS — Bedrock & Vector Stores

Schedule: Week 13

Pune cloud architects in 2026 are expected to design AI features, not just classic three-tier apps. Cover Amazon Bedrock (Claude, Llama, Titan, Mistral, Stable Diffusion models on a single API), Bedrock Guardrails for content filtering, Bedrock Knowledge Bases backed by Amazon OpenSearch Serverless or Aurora pgvector, and Bedrock Agents for tool-using assistants. Build a small retrieval-augmented generation (RAG) service — S3 PDFs → Knowledge Base → Bedrock model → API Gateway endpoint — that you can demo end-to-end in interviews.

Topics

- Amazon Bedrock — model catalog, on-demand vs provisioned
- Bedrock Guardrails for safety filtering
- Vector storage — OpenSearch Serverless and Aurora pgvector
- Bedrock Knowledge Bases for managed RAG
- Bedrock Agents — tool calling, action groups
- SageMaker AI overview for custom training
- Cost and latency trade-offs across models

Module 16: Capstone Project & SAA-C03 Exam Preparation

Schedule: Weeks 13–14 + 2 weeks placement prep

Two weeks of full-time capstone work plus structured Solutions Architect Associate (SAA-C03) preparation. Pick one of three capstone architectures (see Capstone Projects). For exam prep we run question-bank drills, scenario walkthroughs, and two full-length mock exams under timed conditions — students typically score 75–85% on the second mock if they have

done the lab work seriously. Mock interviews target Pune cloud hiring panels (Persistent, BMC, Bajaj Finserv, Synechron). Resume / LinkedIn / GitHub polish included.

Topics

- Capstone implementation, deployment, README
- Architecture review against Well-Architected pillars
- SAA-C03 question-bank drills — 5 domains
- Two full-length timed mock exams
- Cloud-engineer interview prep — scenario design questions
- Resume, LinkedIn, GitHub portfolio polish
- HR mock interview and salary negotiation

Module 17: AI-Assisted Development Workflow

Schedule: Final Week

The skill every 2026 hiring panel now probes for — building real work with AI in the loop, responsibly. Learn to drive AI assistants (GitHub Copilot, Claude, Cursor, and IDE-native AI) to scaffold and accelerate the tools and stack this course covers, generate tests, explain and refactor unfamiliar code, and cut the boilerplate — while keeping you firmly in control of every decision. Heavy focus on guardrails: reviewing each AI suggestion, spotting hallucinated APIs or wrong answers, and handling licensing and data-privacy concerns. Close with a mini-project that takes a deliverable end-to-end using an AI-assisted workflow, then fold the same tooling into version control and everyday team practice.

Topics

- AI assistants — GitHub Copilot, Claude, Cursor, IDE-native AI
- Effective prompting for this course's stack — scaffolding, boilerplate, config
- AI-assisted test generation and coverage
- Explaining, refactoring, and modernising unfamiliar code with AI
- AI-driven review, error detection, and quality checks
- Generating and maintaining documentation with AI
- AI debugging — interpreting errors, logs, and failing output
- Guardrails — reviewing output, avoiding hallucinations, licensing & data privacy
- Team workflow — AI in the editor, in reviews, and in delivery pipelines
- Mini-project — a deliverable built end-to-end with an AI-assisted workflow

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.